

An Explainable Evidence Linking Framework for Automated Fact Verification

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Abstract—Triple-R is a new fact-checking method that uses artificial intelligence and retrieval-based evidence to fight misinformation on internet platforms. It has three main parts: Retriever, Ranker, and Reasoner. These components offer practical insights and contextual explanations. Okapi BM25 is designed to retrieve information from external web source like Wikipedia. The ranker module combines BM25 and Sentence BERT, while the Llama 3.2 is chosen as reasoner module. An innovation in ranker and reasoner module is shown, which shows an accuracy of 56%. The proposed method seeks to increase fact-checking systems' interpretability and openness by offering clear justifications for decision-making procedures. The proposed hybrid ranking method is developed and effectiveness of the approach is tested on a dataset, which is accessed from <https://www.unb.ca/cic/datasets/index.html>.

Index Terms—Misinformation Detection, Large Language Models, Natural Language Processing, BM25, SBERT, Llama3.2, Triple-R Dataset, LIAR Dataset, Automated Fact-Checking

I. INTRODUCTION

Spreading misinformation on social media has become a major problem. This has created a need for fact-checking techniques. Traditional methods often struggle when it comes to scaling and bias. This gap has led to the use of technologies like Artificial Intelligence and Natural Language Processing. The GPT-3.5-Turbo model, along with these technologies, has helped develop AI-based fact-checking frameworks. These frameworks can quickly retrieve, rank, and verify claims, improving the accuracy of unbiased reasoning. Triple-R is a new AI fact-checking pipeline proposed by Mohammadamin Kanaani and others to fight misinformation on social media [1]. This model expanded the LIAR dataset by adding external evidence and AI-generated reasons. With Explainable AI principles, the model's decision-making is transparent and easy to understand. This also helps verify media credibility for regulatory compliance, making it a strong tool against misinformation. Its performance analysis shows its potential for researchers and practitioners, encouraging effective responses to misinformation on social media.

Research on automated fact-checking has found several limitations and errors in different methods. A major issue is that many systems focus only on text and ignore the importance of multimodal information, especially images, when verifying

claims. Additionally, many systems do not provide clear explanations, which can lower trust [2]. The fixed nature of parametric knowledge in Large Language Models (LLMs) also results in poorer performance over time. A significant gap in current research is the lack of multimodal datasets that can handle both claims and evidence at the same time. Many systems focus on single aspects of fact-checking, like retrieving evidence or summarizing information, instead of providing a complete solution. Understanding LLM outputs remains difficult, and some methods produce unclear or aimless prompts. Assessing fact-checking systems is tough due to time-related factors, outdated truths in datasets, and the reliability of external evidence found through web searches [4]. The high computational cost of advanced LLMs and the reliability of web scraping also add practical challenges. Common problems include "premature judgment" and "failed evidence retrieval" by models [5]. Automatically improving prompts for LLMs struggles with navigating the large space of possible prompts. Methods can be directionless or rely on discrete manipulations without clear semantic direction, leading to suboptimal prompts or increased computational costs [6]. The objective of the paper is to develop a Triple-R inspired framework which can retrieve the evidence similar to the claim, rank them based on their similarity and eventually provide the reasoning, supporting the claim.

II. RELATED WORK

Kanaani et al.'s Triple-R framework is a new approach that combines three components: a retriever, a ranker, and a reasoner, to improve the accuracy and explainability of fact verification processes. This approach aims to enhance transparency by providing human-readable explanations for the decisions made by the system. However, it also faces issues such as excessive computational cost, possible evidence bias, and accuracy issues. The work includes various methods for automated fact-checking, highlighting their merits and demerits. Despite these challenges, the Triple-R framework offers a promising solution for improving the verification process [1]. Alhindi and Petridis's LIAR-PLUS dataset [2] progresses the existing LIAR dataset further by expanding the material to cover automatically generated reasons specifically for fact-checking claims made in various contexts. The authors

were able to significantly demonstrate how the model of justification, that are human-supplied can significantly enhance the ability to appropriately predict the truth or veracity of various claims. Furthermore, it is significant to incorporate temporal and numeric information when modeling the claim and its corresponding justification because such incorporation would be crucial in appropriately capturing and knowing the correct context of a particular statement.

DEFAME, which has been presented by Braun et al., is a multimodal fact-checking system that deals with both images and text to prove or disprove claims, solving the disadvantage of text-based systems. DEFAME combines many of the developments for automated fact-checking into one system and dynamically retrieves evidence and provides exhaustive reports. DEFAME confronts misinformation with multimodal arguments and evidence and dynamically pulls in evidence as required. DEFAME's search-engine dependence brings with it the danger of including unreliable information [4].

Ahmad et al. suggested a machine learning framework for fake news detection using ensemble methods like Random Forest, Gradient Boosting, and AdaBoost. The study highlighted the advantages of using multiple classifiers to improve accuracy and minimize overfitting. However, this approach had some drawbacks. It did not compare with deep learning models, did not examine feature interpretability or the effects of specific linguistic cues, and used small datasets. The paper also did not tackle issues such as concept drift, adversarial inputs, and the needs for real-time detection. Future research should concentrate on scalability, interpretability, and resilience in real-world situations [6].

Golovin et al. suggested a hybrid recommender system for detecting fake news using knowledge graphs and Graph Attention Networks (GATs). The system includes graph nodes with attention weights. It detects influential users and structural relationships. The system achieved a high F1-score of 95 percent on the FakeNewsNet dataset. It surpassed traditional machine learning and transformer-based models. However, the model's reliance on social propagation data limited its effectiveness for newly posted or rarely shared news. It also did not tackle adversarial robustness or resistance to changed graph structures. The system only handled unimodal data and did not consider multimodal inputs, such as images or behavioral patterns [8].

Mishra et al.'s FACTIFY is a large-scale multimodal fact-verification dataset that includes 50,000 individual points from India and the United States. This dataset is different from most existing research on fact-checking, which often relies on unimodal approaches and synthetic data generation. The main objective of FACTIFY is to address the lack of focus on fact verification across multiple modalities. Although FACTIFY can classify claims into binary labels, it acknowledges that some claims may be in a grey area and partially true or false, rather than easily categorized into a simple true or false label [10].

The rise of misinformation and disinformation makes strong fact-checking systems essential. However, traditional LLM-based and multimodal pipelines often struggle with serious computing challenges.. This problem is especially important for edge devices or places with limited resources. They need efficient models to work well. This computing problem highlights the need for lightweight, hybrid models that maintain high accuracy while significantly lowering operational costs. Current systems, such as Triple-R, provide textual justifications, but the wider field of multimodal explainability, especially in grasping the complex connections between images and text, is still largely untested and underdeveloped.. Additionally, the lack of standard methods for creating and assessing explanations across different types and designs hinders consistency, comparability, and replicability in explainable fact-checking systems. This gap points to an important research opportunity to find new methods that not only enhance computing efficiency but also improve the understanding of complex multimodal information. This paper suggests a new approach to tackle these issues by creating a lightweight, hybrid fact-checking framework. This framework will combine multimodal large language models with specialized, efficient modules for better explainability and lower inference costs.

III. METHODOLOGY

The fact-checking system described in this study is similar to the Triple-R, an AI fact-checking system with three stages: the Reasoner, Ranker, and Retriever. The method is a modular pipeline coded in Python that combines deep learning-based natural language processing techniques with traditional information retrieval. The LIAR dataset originally contained 12,836 statements with truth labels. The Triple-R dataset builds on this as a resource for fact-checking. Each claim's supporting evidence comes from the public domain, is ranked for relevance, and then a large language model generates a truthfulness label and a detailed explanation. This approach makes AI-driven fact-checking clear and understandable. The dataset includes about 10,269 claims for training, 1,280 for testing and evaluation, and 1,287 for development. It is split into standard subsets for developing models and ensuring fair assessment. This approach maintains the original process of the LIAR dataset and retains all class distributions.

A. Practical Workflow

- 1) Dataset : includes preliminary assertions or statistics that need to be confirmed. Wikipedia is the main source of information. Additional Open Domain Sources for Channels of Information (e.g., public databases, news).
- 2) Retriever Module (BM25) : uses keyword matching to find and filter pertinent documents or evidence segments.
- 3) Ranker Module (BM25 + SBERT):for results that are topically relevant, the obtained evidence is ranked using both conventional keyword approaches and semantic similarity [3].
- 4) Reasoner Module (LLAMA3.2): makes use of ranking evidence to determine the validity of the claim. labels

are given according to a clear judgment (e.g., true, false, etc.).

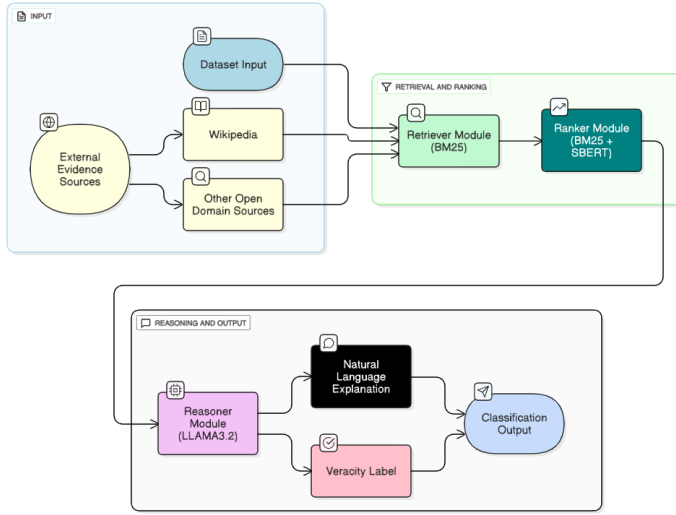


Fig. 1. Block diagram of the proposed Triple-R fact verification system.

B. Dataset Loading and Preprocessing

A JSON-formatted dataset of textual claims and labels is loaded at the beginning of the operation. The file structure is verified and usable records are provided via a custom load dataset method. An identifier, a ground-truth label, and a plain language statement are all included in each claim record [1].

C. Evidence Retrieval using BM25

The pipeline's initial part, the Retriever, uses the Wikipedia API to obtain textual support for a claim. Brief summaries are retrieved by the procedure according to the substance of the claim. Paragraphs are ranked by the BM25 OKAPI algorithm according to lexical closeness to the claim, word frequency, and inverse document frequency. After that, these paragraphs are forwarded for additional screening [1]. The OkapiBM25 version brings some innovation to the TF-IDF formula starting from the Inverse Document Frequency [3]. The main difference is that the IDF in the OkapiBM25 takes into account frequency of terms and penalizes terms that are common:

$$IDF(q_i) = \ln \left(1 + \frac{docCount - f(q_i) + 0.5}{f(q_i) + 0.5} \right) \quad (1)$$

where docCount is the total number of documents and $f(q_i)$ is the number of documents containing term q_i . The Term Frequency part of the formula becomes:

$$TF(q, d) = \frac{f(q_i, d) \cdot (k_1 + 1)}{f(q_i, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{avgdl} \right)} \quad (2)$$

where $f(q_i, d)$ is the term frequency of the term q_i in document d , $avgdl$ is the average document length in the collection and

k_1 and b are free parameters, for this project the default values have been chosen as $k_1 = 1.2$ and $b = 0.75$ [3].

BM25 term-weighting and document-scoring function

$$\frac{tf}{k_1 \left((1 - b) + b \frac{dl}{avgdl} \right) + tf} \cdot w_i^{RSJ} \quad (3)$$

RSJ in Wi(RSJ) stands for Robertson-Sparck Jones Weighting [11].

The term weight for each word is given by

$$w_i^{IDF} = \log \frac{N - n_i + 0.5}{n_i + 0.5} \quad (4)$$

$$B = \left((1 - b) + b \frac{dl}{avgdl} \right), \quad 0 \leq b \leq 1 \quad (5)$$

The parameter b controls document length normalization. $b = 1$ will apply full document-length normalization, while $b = 0$ will ignore the length factor [7].

$$score(D, Q) = \sum_{i=1}^n IDF(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{avgdl} \right)} \quad (6)$$

In the equation 6, $IDF(q_i)$ represents the inverse document frequency, n is the total number of documents, and n_i is the number of documents that contain the term q_i , soft length normalization component. Here, dl is current document length and $avgdl$ is average document length across all documents in the collection. k_1 called as saturation parameter controls the impact of Term Frequency(TF) on the document score. If $k_1 = 0$, only IDF value will be considered for document score. The paper suggests that setting $1.2 < k_1 < 2$ generally yields good results [7].

D. Hybrid Evidence Ranking using BM25 and SBERT

To address the inherent limitation of lexical-only retrieval, the Ranker uses a hybrid ranking model to improve the relevance of selected evidence [1]. The model uses BM25 to rank and select the top 10 candidate evidence snippets, which are processed using Sentence-BERT. The candidate and claim paragraphs are then inserted into dense vector representations. Semantic similarity is determined using cosine distance, resulting in six semantically appropriate paragraphs for final reasoning. This combination method leverages both superficial token matching and deeper semantic understanding and significantly enhances the relevance of selected evidence [1]. BM25 and SBERT calculates relevance in fundamentally different ways. Their mathematical relationship is not direct, but they are often combined in practice for a more effective document ranking. The BM25 score between a query Q and a document D is:

$$\text{BM25}(Q, D) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)} \quad (7)$$

$f(q_i, D)$: frequency of term q_i in document D

$|D|$: length of document D

avgdl: average document length in the corpus

k_1, b : hyperparameters

$\text{IDF}(q_i)$: inverse document frequency of term q_i

BM25 generates a non-normalized score, generally unbounded, representing term overlap and presence [13]. SBERT generates dense vector embeddings for the query and document. The cosine similarity is given by:

$$\text{cosine_sim}(\vec{q}, \vec{d}) = \frac{\vec{q} \cdot \vec{d}}{\|\vec{q}\| \|\vec{d}\|} \quad (8)$$

where $\vec{q}$ and $\vec{d}$ are the embedding vectors of the query and document, respectively [14]. This value ranges from -1 to $+1$ (typically 0 to 1 in semantic search), representing the semantic closeness of the query and document.

There is no intrinsic mathematical mapping between BM25 scores and SBERT cosine similarity, since they represent two independent notions of relevance:

- BM25: Measures keyword overlap and term frequency.
- Cosine Similarity: Measures semantic proximity in embedding space.

Combination of these scores [15] is typically achieved by normalization and weighted fusion:

$$\text{CombinedScore} = \alpha \cdot \text{Norm. BM25} + (1 - \alpha) \cdot \text{Cos Similarity} \quad (9)$$

where α is a weighting parameter (often determined empirically) [16]. BM25 scores are normalized before combining, often using min-max scaling:

$$\text{BM25}_{\text{norm}}(d) = \frac{\text{BM25}(d) - \min(\text{BM25})}{\max(\text{BM25}) - \min(\text{BM25})} \quad (10)$$

This brings BM25 into a comparable range, similar to cosine similarity [15].

E. Reasoning and Claim Classification using LLaMA 3.2

The Reasoner, the final pipeline component, uses the LLaMA 3.2 large language model to reason over textual evidence in automated fact-checking workflows. The model consists of a structured prompt with the input claim and top six ranked evidence snippets, analyzing the evidence, generating an interpretable reason paragraph, and output a final truth label [8]. The model can output multiple verdicts, reflecting the complexity of real-world information verification tasks. Instead of just "true" or "false," the model can output several distinct verdicts such as "mostly true," "half true," "barely true," or "pants-on-fire", reflecting the complexity found in

real-world information verification tasks [9]. In order to evaluate the correctness of the claim, the reasoning process uses a chain of thought reasoning, which divides verification into more manageable logical steps and is followed by explanation and subsequent questions.

After reasoning, Llama 3.2 assigns a verdict label to the claim. The 6 labels are [9]:

- 1) True: Accurate and completely supported by evidence.
- 2) Mostly True: Largely correct, but minor details missing or needs clarification.
- 3) Half True: Contains both accurate and inaccurate elements.
- 4) Barely True: Only slightly supported; critical facts that would alter interpretation are missing.
- 5) False: Not supported by evidence; factually incorrect.
- 6) Pants-On-Fire: Wildly inaccurate, strongly contradicted by evidence

In some datasets, "Supported", "Refuted", and "Not Enough Information" or similar mappings are used, but the above is a standard six-way distinction in fine-grained fact-checking [12].

F. Label Extraction and Normalization

The extract label method handles the LLM response after inference, retrieving the classification label and confirming that it belongs to the valid set of labels. The label "unknown" is applied by default if none is returned. The output is ready for machine-readable evaluation in this step.

G. Performance Evaluation Metrics

To objectively evaluate the proposed pipeline's performance, a dedicated evaluate function compares predicted labels to the dataset's ground truth values. The evaluation focuses on multiclass classification metrics, which are as follows:

- Accuracy: Total number of correctly classified claims divided by the total number of claims.
- Precision: Weighted average precision of all the label categories.
- Recall: Ratio of true positive predictions to all positive instances present.
- F1-score: Harmonic mean of recall and precision.

These are computed using the sklearn.metrics module, and initial testing was conducted on a random sample of 1,283 claims. The proposed framework for fact checking was written in Python 3.10 and tested in a local development environment. The BM25 algorithm retrieves relevant text from web sources such as Wikipedia. BM25 and sentence transformers use semantic similarity to rank sentences. The Ollama API is communicated with via HTTP, and the scikit-learn packages are utilized for reasoning.

IV. RESULTS AND DISCUSSION

The efficiency of the Triple-R fact-checking process was confirmed and evaluated using a publicly available dataset

of 1,283 assertions. The system’s performance was assessed using conventional multi-class classification measures such as Accuracy, Precision, Recall, and F1-score, which are widely used in NLP-based classification tasks. The evaluate function is used to compare the system output with the ground truth labels. The metrics show that the model can categorize factual statements into six groups: Pants on Fire, True, Mostly True, Half True, Barely True, and False. Although the performance is moderate, it is consistent with what might be anticipated from early implementations of open-domain retrieval-based hybrid evidence-based reasoning pipelines.

A. Analysis of Prediction Behavior

The proposed system performed rather well when the claims were accurately reflected in Wikipedia or when the evidence obtained had a significant contextual overlap with the claim, according to a manual analysis of several test instances. Lexical matching alone would have misranked semantically comparable information, however the hybrid ranking method of BM25 + SBERT worked well for fine-grained evidence snippet refinement.

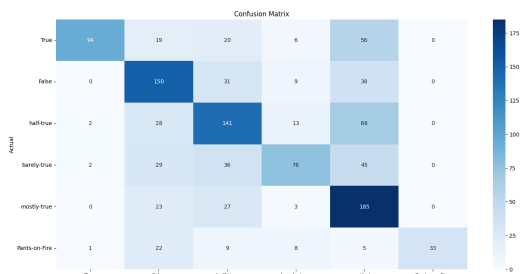


Fig. 2. Confusion matrix.

The Fig.2 shows the model’s accurate predictions are shown on the diagonal, with Mostly-True (185), False (150), and Half-True (141) being the most frequently and accurately predicted. The model’s weakest predictions are on Pants-on-Fire as 22 were misclassified as False and 9 as Half-True and 8 as Barely-True. Such misclassifications reveal the model’s challenges in separating the most extreme claims. The model has an even clearer problem with discerning the difference between Half-True and Barely-True, with 28 False claims being incorrectly classified as Half-True. With respect to the claims, the model has demonstrated the most competency in classifying True and Mostly-True claims. For Pants-on-Fire claims, the model needs to become more precise and simultaneously deal better with the more subtle truth claims. The model’s accuracy rate is 57, indicating that there is good baseline performance on the model, but there are still challenging areas that need to be worked on.

The Fig. 3 shows the distribution of top BM25 scores for 1,200 claims, separating correctly predicted claims (green) from incorrectly predicted ones (red). The y-axis shows the BM25 score of the top-ranked retrieved evidence, while the x-axis lists individual sample indices. A dotted blue line

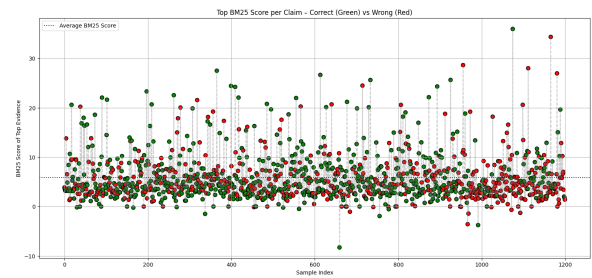


Fig. 3. BM25 scores per claim.

marks the average BM25 score across all claims, providing a baseline for comparison. Generally, correct predictions (green) gather around or above this average line. This suggests that higher BM25 scores are linked to more relevant evidence retrieved and improved model accuracy. In contrast, incorrect predictions (red) often fall below or near the average line, indicating less relevant evidence. The scattered distribution of both classes shows differences in retrieval success, but the visible trend confirms that stronger lexical similarity (higher BM25 scores) improves factual alignment and retrieval quality.

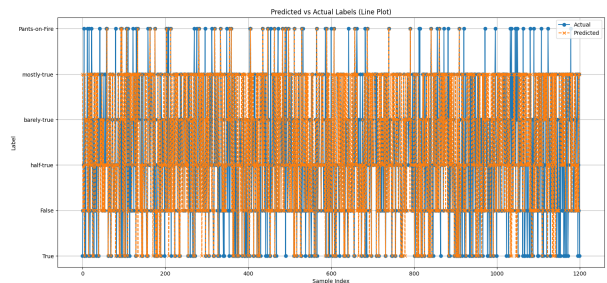


Fig. 4. Predicted versus Actual labels.

Fig.4 shows the predicted and observed labels for fact-checking in 1,200 samples. The x-axis corresponds to the sample index and the y-axis presents the comparison of truth-relevance categories: True, False, Half-true, Barely-true, Mostly-true, and Pants-on-Fire . The blue line represents the actual fact-checking labels, the orange dashed line indicating the predicted labels for those samples. When the blue line overlaps the orange dashed line, it indicates the sample was correctly categorized; when the two lines do not overlap, the sample was miscategorized. The dense compaction of the lines demonstrates that it is difficult to disambiguate samples with different levels of truths. The overlaps and mismatches can help identify scenarios where the model’s logic or evidence-based approach needs to be improved. Therefore, the accuracy obtained for this case was 57%.

However, when it comes to handling assertions that lack sufficient evidence or are unclear, the rational model (LLaMA 3.2) is inconsistent. In other cases, the model misclassified because it fell back on general thinking. The reason engine may require quick optimization or fine-tuning, as the most fre-

TABLE I
COMPARISON WITH OTHER WORK

Technique	Accuracy	Precision	Recall	F1-Score
Proposed	57	64	57	57
Llama2(7b)	42.72	51	42	40
GPT-3.5-Turbo	37.1	36.16	34.35	30.3

quent misclassifications happened between semantically comparable labels. Testing found two major limitations: - Variation in evidence quality on Wikipedia, particularly for expert or event-oriented statements. The retriever struggled when there was insufficient Wikipedia coverage. The second constraint is label ambiguity. Because truth labels such as Half True and Barely True are subjective, annotators may disagree. The ambiguity in the labeling aids in both training and judgment accuracy. Table I shows a comparison of the proposed fact checking framework with some other works, where Llama2 (7b) and GPT-3.5-Turbo is used as reasoner [1].

V. CONCLUSION

Triple-R demonstrates the feasibility of automating fact-checking by combining information retrieval and deep learning techniques. The hybrid ranking system makes evidence more relevant, while the Llama-3.2 model provides straightforward reasoning. However, difficulties such as the quantity of processing required, bias in evidence selection, and reasoning correctness remain to be addressed. In the future, academics may look into larger datasets, improved ranking models, and more solid reasoning frameworks. Future research will focus on several key areas like multimodal integration with attentional Fusion, veracity Nuance Modeling, real-time adaptability via active Learning and Xplainable AI (XAI) for Transparent Reasoning. Through incorporation of these elements, a new-generation fact-checking system is better able to deal with the intricacies of contemporary misinformation, providing reliable, transparent, and responsive solutions.

VI. DISCUSSION AND FUTURE WORK

A. Discussion

Using a mix of information retrieval and deep learning, Triple-R shows that it is possible to automate fact-checking. The hybrid ranking method makes evidence more relevant, and the Llama-3.2 model gives clear reasoning. There are still problems to solve, though, such as the amount of computation needed, bias in evidence selection, and accuracy of reasoning. In the future, researchers could look into bigger datasets, better ranking models, and more reliable reasoning frameworks.

B. Future Work

Future research will focus on several key areas:

- 1) Multimodal Integration with Attentional Fusion
- 2) Veracity Nuance Modeling
- 3) Real-time Adaptability via Active Learning
- 4) Source Credibility Evaluation with Blockchain
- 5) Explainable AI (XAI) for Transparent Reasoning

Through incorporation of these elements, a new-generation fact-checking system is better able to deal with the intricacies of contemporary disinformation, providing reliable, transparent, and responsive solutions.

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